AWS Services for Continuous Integration and Continuous Delivery (CI/CD)

Continuous Integration (CI) and Continuous Delivery (CD) are fundamental practices in modern software development, enabling teams to deliver high-quality software rapidly and reliably. AWS provides a comprehensive suite of services designed to support and automate every stage of the CI/CD pipeline, from source code management to deployment and monitoring.

Core AWS CodeSuite Services

AWS offers a set of integrated services, collectively known as the AWS CodeSuite, which form the backbone of many CI/CD implementations on the platform. These services are designed to work together seamlessly, providing end-to-end automation for the software release process.

AWS CodePipeline serves as the orchestration engine for CI/CD workflows. It allows users to model, visualize, and automate the steps required to release software. A pipeline typically includes stages such as source, build, test, and deploy, integrating with various AWS services and third-party tools to move code changes through the release process. This ensures rapid and reliable delivery of features and updates by automating the entire release cycle.

AWS CodeBuild is a fully managed continuous integration service that compiles source code, runs tests, and produces deployable software packages. It eliminates the need to provision, manage, and scale build servers, allowing developers to focus on writing code. CodeBuild scales automatically to meet demand, processing multiple builds concurrently.

AWS CodeDeploy automates software deployments to a variety of compute services, including Amazon EC2 instances, AWS Fargate, AWS Lambda, and on-premises servers. It streamlines the deployment process, reducing the risk of manual errors and ensuring consistent application deployments. CodeDeploy supports various deployment strategies, such as in-place and blue/green deployments, to minimize downtime during updates.

AWS CodeCommit is a secure, highly scalable, managed source control service that hosts private Git repositories. It eliminates the need to operate and maintain a separate source control system, integrating directly with other AWS services. While there was a period in 2024 where AWS de-emphasized CodeCommit, it has since returned to General Availability in late 2025, indicating its continued role in the AWS ecosystem.

AWS CodeArtifact functions as a fully managed artifact repository service. It simplifies the process of storing, publishing, and sharing software packages and dependencies, supporting popular package managers like npm, PyPI, Maven, and NuGet. CodeArtifact ensures that development teams have consistent and secure access to their software components throughout the CI/CD pipeline.

Specialized CI/CD Services

Beyond the core CodeSuite, AWS provides specialized services that offer streamlined CI/CD capabilities for specific application types or development paradigms.

AWS Amplify offers a comprehensive platform for building, deploying, and hosting full-stack web and mobile applications. It includes a fully managed CI/CD service that provides a zero-configuration deployment experience for single-page applications (SPAs) and server-side rendered (SSR) applications. Amplify automates the entire development workflow, from code changes to production deployment.

AWS App Runner is a fully managed service that makes it easy to deploy containerized web applications and APIs at scale. It provides a direct CI/CD workflow, allowing developers to connect their GitHub repositories or Amazon Elastic Container Registry (ECR) images. App Runner automatically builds, deploys, and scales the application, abstracting away infrastructure management.

AWS Proton is a managed delivery service designed for platform engineering teams. It enables the creation and management of standardized infrastructure templates and self-service deployment tools for developers. Proton helps enforce architectural best practices and consistency across an organization’s development environments, integrating CI/CD pipelines into its service templates.

Supporting Services for CI/CD

Several other AWS services play crucial roles in enhancing and supporting CI/CD pipelines, providing capabilities for infrastructure management, security, and automation.

Amazon Elastic Container Registry (ECR) is a fully managed Docker container registry that makes it easy to store, manage, and deploy Docker container images. ECR integrates with other AWS services like CodeBuild and App Runner, serving as a central repository for container images used throughout the CI/CD process.

AWS CloudFormation and **AWS Cloud Development Kit (CDK)** are powerful Infrastructure as Code (IaC) tools. CloudFormation allows users to define AWS resources in declarative templates, while CDK enables defining infrastructure using familiar programming languages. Both are instrumental in automating the provisioning and management of CI/CD pipelines and the underlying application infrastructure, ensuring consistency and repeatability.

AWS Lambda is a serverless compute service that allows running code without provisioning or managing servers. It is frequently used within CI/CD pipelines for custom automation tasks, such as triggering builds, performing pre-deployment validations, sending notifications, or integrating with external systems.

AWS Step Functions provides a serverless workflow service for orchestrating complex, distributed applications and microservices. It can be used to manage intricate deployment workflows that go beyond the capabilities of CodePipeline, allowing for sequential, parallel, branching, and error-handling logic.

AWS Systems Manager (SSM) offers a suite of capabilities for managing AWS resources. Within CI/CD, its Parameter Store can be used to securely store configuration data and secrets, while AppConfig helps manage and deploy application configurations independently of code deployments.

AWS Secrets Manager is a service that helps protect access to applications, services, and IT resources. It enables easy rotation, management, and retrieval of database credentials, API keys, and other secrets throughout their lifecycle, ensuring that sensitive information used in CI/CD pipelines is securely handled.

Conclusion

AWS provides a robust and extensive ecosystem of services for building and managing CI/CD pipelines. From core services like CodePipeline, CodeBuild, CodeDeploy, CodeCommit, and CodeArtifact to specialized offerings like Amplify, App Runner, and Proton, and supporting services such as ECR, CloudFormation, Lambda, Step Functions, Systems Manager, and Secrets Manager, AWS offers a comprehensive toolkit to automate and optimize the software delivery lifecycle. This enables organizations to achieve faster release cycles, improve software quality, and enhance operational efficiency.